GEK-45150A



GE Drive Systems

INSTRUCTIONS

531X134EPRB_G1
ENCODER PROCESS INTERFACE CARD

Renewal Part

IMPORTANT
INFORMATION

CAUTION: To ensure proper operation of the motor drive controller, these instructions must be followed for proper set-up and installation of this replacement card.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

SAFETY PRECAUTIONS

WARNING - SHOCK AND BURN HAZARD: Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.

CAUTION: Treat all cards with static sensitive handling techniques. Use "grounding" strap when changing cards and store cards in the anti-static bags they are shipped in.

NOTE: Read all WARNINGS, CAUTIONS and NOTES on the pertinent drive instruction Book (GEK) prior to removing or installing any card.

CARD REPLACEMENT PROCEDURE

1. Carefully disconnect all cables.
2. Remove the four screws securing the Encoder Process card (Figure 1) to the standoffs on the Main Control card and remove card.
3. Set all jumpers on the replacement card exactly as the jumpers on the failed card except where instructed otherwise by this bulletin.
4. Install the new card by reinserting and tightening the four screws removed in step 2.
5. Reconnect all cables verifying that they are properly seated at both ends.
6. Set all potentiometers in the same physical position as the potentiometers on the failed card and fine tune as required per instructions in the Adjustments Section of the Drive instruction Book.

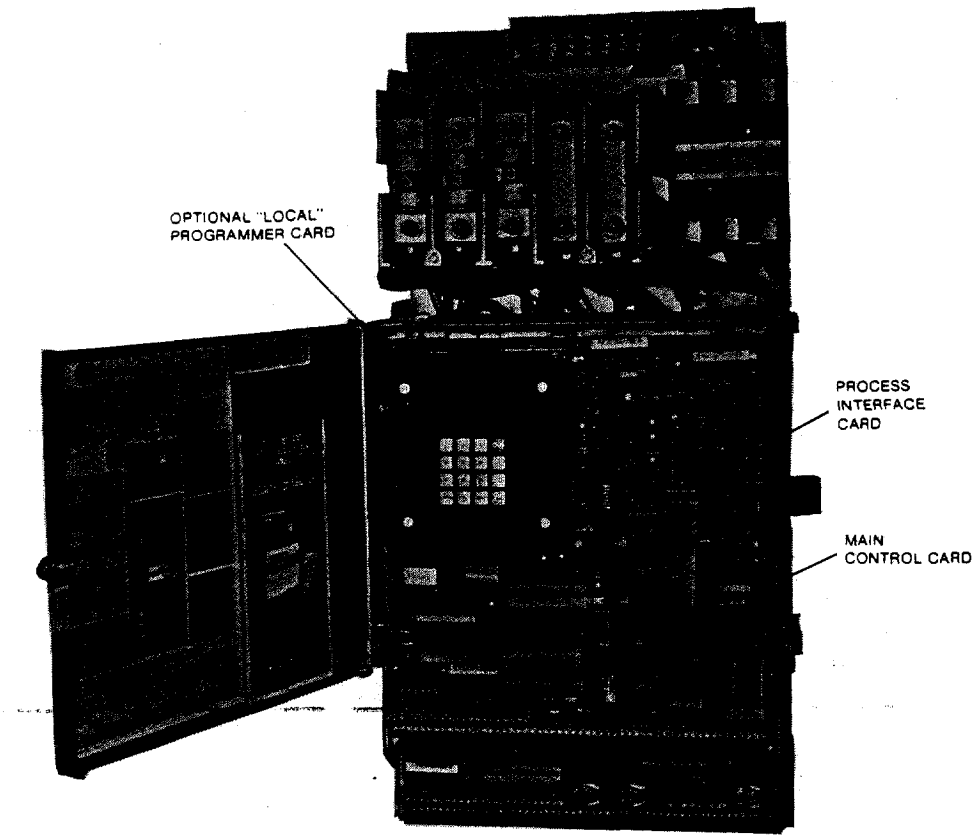


Figure 1. ENCODER PROCESS INTERFACE CARD

JUMPER CONFIGURATION

This circuit card is a functional replacement for the cards specified in the tables provided in these instructions. In order to incorporate design enhancements while continuing to provide compatibilty with a variety of previous designs, changes in card layout (including the relocation, addition or deletion of certain jumpers) may have been required. Jumper locations are shown in Figure 2. Using the jumper listings provided with the drive (refer to the **CUSTOM INSTRUCTION BOOK**) and the card which is being replaced as guides, configure the jumpers of the new card as follows:

1. Locate the card numbers for the old and replacement cards by referring to the appropriate table.
   -- The part number for the card is printed on a white label that is affixed to the card and begins with the number "531X". This is NOT the number "F531X_ _ _ which is silkscreened directly on the circuit board.
2. Cross reference between the card number and configuration jumpers per TABLE 2. Entries indicate the following:
   Y Jumper is provided
   N Jumper is NOT provided
   1,2,3 Numerical reference to NOTES
3. Set jumpers listed as provided on both cards in the same position on the replacement card as they were on the card being replaced, (unless instructed otherwise in the notes provided with the table).
   -- Other jumpers should be set as indicated in the referenced notes.
   -- See TABLE 1 for a functional description of the jumpers.
4. Record any jumper setting changes on the Jumper Listing provided with the drive (located in the drive door pocket with the Parameter List).

IMPORTANT: Certain versions of cards may have a part number suffix such as "S" or "H", instead of "G". These cards are functionally identical to the "G" version. When ordering spare or replacement cards, be sure to use the part number on the card or on the Renewal Parts List provided with the drive.

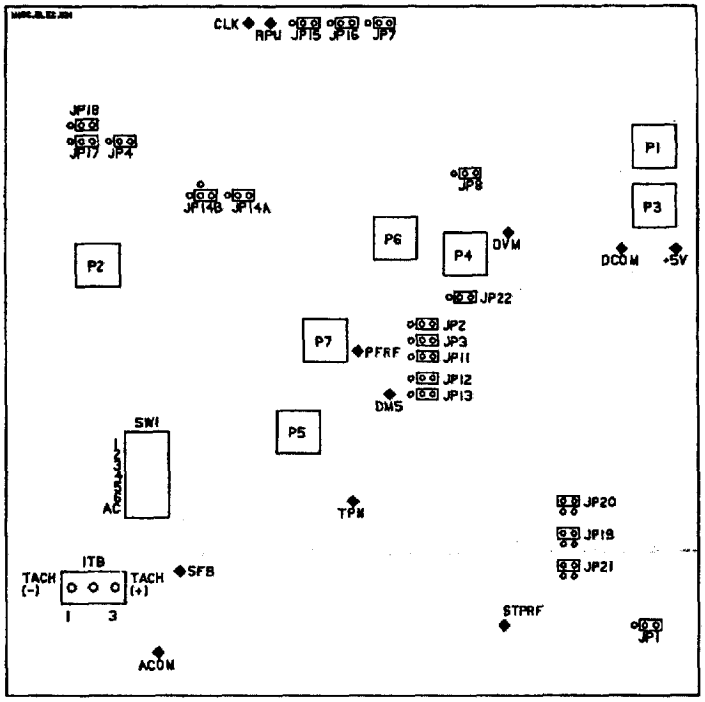


Figure 2. ENCODER PROCESS INTERFACE CARD LAYOUT DIAGRAM

TABLE 1: JUMPER DESCRIPTION - 531X134EPRB_G1 CARDS

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional)	Jumper	JP1		RXD Mode Select **1-2:** Enables RS422 reception on the serial link data line. **2-3:** Enables RS232C reception.
		JP2		5-50 Milliamp Select (see JP3)
		JP3		4-20 Milliamp Select - JP2 and JP3 select the process follower mode per the following table: **JP2 JP3 Range** **1-2: 1-2:** 5-65ma current loop follower **1-2: 2-3:** 5-50ma current loop follower **2-3: 1-2:** 4-20ma current loop follower **2-3: 2-3:** 1-5ma current loop follower or 10V voltage follower
		JP4		SMET Type Select **1-2:** Unipolar speed meter driver mode **2-3:** Bipolar meter driver mode (DC tach only)
		JP5		Marker Pulse Mode Select **1-2:** Differential RS422 compatible, with 0.33 ohm termination **2-3:** Single ended input at PCP
		JP7		Marker Pulse Enable **1-2:** Not used **2-3:** Enables marker pulse to TUP
		JP8		Analog Tach Filter Select **1-2:** Enables 4ms filter in analog tach feedback; required for AN tach operation **2-3:** Disables 4ms filter; required for speed regulator performance above 20 radians
		JP9 & JP10		Digital Tach or Encoder Mode Select **1-2:** Differential RS422 compatable with termination resistor **2-3:** Single-ended input at 10PL-5 and 10PL-7 **Open:** Differential operation with no terminating resistor.
		JP11		TXD Mode Select **1-2:** 12PL-9, 12PL-12 RS422 transmitted datat mode **2-3:** 12PL-9 RS232 transmitted data mode
		JP12		DTR Mode Select **1-2:** 12PL-8, 12PL-13 RS422 DTR handshake mode **2-3:** 12PL-8 RS232 DTR handshake mode

Location	Jumper or Pot	Number	Nomenclature	Description
Encoder Process Interface Card (optional) (continued)	Jumper	JP13		DTR RS232 Logic Polarity **1-2:** 12PL-8 high=drive ready to receive **2-3:** 12PL-8 low=drive ready to receive
		JP14		CTS Mode Select **1-2:** For RS422 CTS handshaking **2-3:** For RS232 CTS handshaking **NOTE:** *For Series 6 interfaces using the CCM card, set JP1 and JP11 to 1-2 and JP12 and JP13 to 2-3. 12PL-7 should be tied to +5V.*
		JP15		Encoder Counter Direction Control **1-2:** Normal control by encoder direction of rotation. **2-3:** Up only
		JP16		Encoder A Quad B Enable **1-2:** Normal A quad B encoder operation **2-3:** Count on leading edge of track A only
		JP17		LUP Marker Pulse Select **1-2:** Marker pulse not connected to 9PL-14 (normal operation) **2-3:** Marker pulse circuit output passed to 9PL-14 for use by LUP
		JP18		LUP DM5 Select **1-2:** DM5 not connected to 9PL-13 (normal operation) **2-3:** DM5 connected to 9PL-13 for use by LUP

TABLE 2: 531X134EPRB_G1 CARD HARD JUMPER MATRIX

Jumper Group	1	2	3	4	5	6	7	8	9	10	11	12
AAG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
ABG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
ACG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
BAG1	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y
BBG1	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y
BCG1	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y
BDG1	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	Y
BEG1	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	Y
BFG1	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	Y
BGG1	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	Y
BHG1	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	Y

TABLE 2: 531X134EPRB_G1 CARD HARD JUMPER MATRIX (contd.)

Jumper Group	13	14	14A	14B	15	16	17	18	19	20	21	22
AAG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
ABG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
ACG1	All AA, AB AND AC BOARDS HAVE BEEN FIELD RETROFITTED.											
BAG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BBG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BCG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BDG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BEG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BFG1	Y	Y	N	N	Y	Y	Y	Y	1	1	1	2
BGG1	Y	N	3	3	Y	Y	Y	Y	1	1	1	2
BHG1	Y	N	3	3	Y	Y	Y	Y	1	1	1	2

Y	=	Jumper present, set same on new board as old board
N	=	Jumper NOT present
1	=	See Note 1 for proper jumper settings
2	=	JP22 in the 1-2 position always
3	=	See NOTE 2 for jumper setting

NOTE 1: JP19, JP20 and JP21 functionally replace JP9, JP10 and JP5 respectively and should be set as follows:

MAX INPUT VOLTAGE	JP19 DTA-/DTA [3TB63, 62]	JP20 DTB-/DTB [3TB65,64]	JP21 DTM-/DTM [3TB60,58]
1.5V	3-4:	3-4:	3-4:
5V	2-3:	2-3:	2-3:
10V	1-2:	1-2:	1-2:
15V	1-4:	1-4:	1-4:

NOTE 2: JP14A and JP14B functionally replace JP14 and should be set as follows:

Original JP14 Position	New JP14A Position	New JP14B Position	Connection Description
1-2:	1-2:	1-2:	CTSA Enabled CTSB Enabled (differential inputs)
2-3:	1-2:	2-3:	CTSA Enabled CTSB @ D Common (single ended input)
2-3:	2-3:	2-4:	CTSA @ D Common CTSB @ -15 Volts (single ended input)
2-3:	2-3:	1-2:	CTSA @ D Common CTSB Enabled (single ended input)

NOTE: *CTSA is 12PL-7 (3TB-32)*
CTSB is 12PL-14 (3TB-41)



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